

Year 2 Assessed Problems

Semester 1

Assessed Problems 7

SOLUTIONS TO BE SUBMITTED

ON CANVAS BY

Wednesday 26 November

at 17:00

Quantum Approach to Solids - Assessed Problem 1

1. Iron

At room temperature, iron is a metal with a body-centered cubic crystal structure, with one atom per lattice site, and a cubic lattice parameter of $2.86 \text{ \AA}$.

- (a) Draw a labelled sketch of the cubic unit cell of iron. [2]
- (b) The packing fraction of a material is the percentage of the total volume of a solid occupied by atoms. Calculate the packing fraction for iron, assuming that the iron atoms act as hard spheres that are in contact with each other. [2]
- (c) Draw several repeats of the unit cell and sketch on the $(1, \bar{1}, 0)$ plane. [1]

Quantum Approach to Solids - Assessed Problem 2

2. Polyacetylene

Polyacetylene is a long chain polymer, of the form $(-\text{CH}=\text{CH}-\text{CH}=\text{CH}-)$ where ‘-’ is a single bond and ‘=’ is a double bond. Longitudinal vibrations parallel to the main chain can be calculated by modelling the polymer as a 1-D chain of identical masses M , representing a ‘CH’ unit, separated by springs of alternating strength representing the single and double bonds, with force constants K_1 and K_2 .

- (a) Identify the smallest possible unit cell. *[Hint: are the ‘CH’ groups all identical?]*
Write down the (two) equations of motion for the longitudinal displacements of the carbon atoms. [1]
- (b) By use of appropriate trial solutions, show that the characteristic frequencies for these motions along the chain are given by

$$\omega^2 = \frac{K_1 + K_2}{M} \left[1 \pm \left\{ 1 - \frac{4K_1K_2 \sin^2\left(\frac{1}{2}qa\right)}{(K_1 + K_2)^2} \right\}^{\frac{1}{2}} \right]$$

where q is the wavenumber and a is the length of the unit cell. [4]